

MUHAMMAD EHAB SIKANDAR

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Data Driven Professional with strong experience in survey analytics, statistical analysis, and predictive modeling to drive business insights. Skilled in data preprocessing, Excel-based analysis, feature engineering, and Driver analysis. Proficient in Python, SQL, R, and data visualization, with a proven ability to analyze both structured and unstructured data and translate complex findings into clear, actionable recommendations.

EDUCATION

Bachelor of Computer Science | University of Central Punjab | 2024

PROFESSIONAL EXPERIENCE

AI Research Assistant | CodeBricks | Apr 2025 – Present

- Conduct survey analytics on consumer research datasets to extract actionable insights on product preferences, flavor trends, and category performance.
- Supported end-to-end survey analytics projects, handling data cleaning, preprocessing, analysis
- Perform statistical analysis and multivariate techniques (including PLSR, PCA, and Ideal Profiling) in R and Python to identify key drivers of liking and purchase intent, comparing results across tools for robustness.
- Build predictive models and Random Forest analyses to support business decision-making and highlight influential factors in consumer behavior.
- Automate survey data workflows, including preprocessing, cleaning, and structuring of numeric and free-text responses, improving efficiency and data reliability.
- Apply advanced Excel skills to handle large-scale datasets, including formulas, pivot tables, and validation checks for accurate reporting.
- Create charts and visualizations in Power BI, including scatter plots and PCA coordinate plots, to communicate survey trends and insights effectively.
- Utilize SPSS for additional statistical analysis and hypothesis testing as needed for survey data.

Data Science Intern | VentureDive | Jan 2025 – March 2025

FinLLM – AI-Driven Financial Data Analysis

- Contributed to the development of a system enabling a language model (LLM) to query tabular CSV data and extract financial insights.
 - Implemented SQL-like querying using Pandas, allowing comparisons, aggregations, and trend analysis on structured financial datasets.
 - Designed a dataframe-based query workflow for retrieving financial trends directly from CSV files without vector search.
 - Ensured outputs included interpretable reasoning, improving transparency and trust in AI-driven insights.
 - Supported data cleaning, preprocessing, and structuring to ensure accuracy and usability of financial datasets.
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Shelf Sight – Multi-Label Object Detection for Retail Analytics

- Contributed to a retail analytics project by supporting the development of a deep learning–based object detection system using RetinaNet.
- Prepared and structured 1,200+ labeled images in COCO JSON format, ensuring data quality and consistency for model training.
- Assisted in model fine-tuning by adjusting hyperparameters, learning rates, batch sizes, and data augmentation techniques.
- Evaluated model performance using standard classification metrics, supporting model validation and comparison across experiments.

PROJECTS

LibraBot – Intelligent Virtual Library Assistant

- Developed an AI-powered virtual library assistant using a large language model.
- Implemented NLP capabilities to answer user queries, running entirely offline on local hardware.
- Created an interactive avatar using NVIDIA Omniverse and Audio2Face for facial animation and lip-syncing.
- Integrated OCR functionality to extract text from images and retrieve relevant book information.
- Conducted user testing to evaluate system stability, response quality, and usability.
- Deployed the system locally using Flask and a lightweight web interface.
- Utilized libraries such as torch, transformers, bitsandbytes, and others for backend processing.

Sentiment Analysis (Python)

- Implemented sentiment classification using traditional machine learning approaches on text data.
- Performed data cleaning, preprocessing, and feature extraction to improve model performance.
- Evaluated and compared multiple modeling approaches on customer review datasets to assess accuracy and reliability.

House Price Prediction

- Developed regression models to predict house prices using historical housing data.
 - Applied feature engineering techniques, including handling missing values, normalization, and encoding categorical variables.
 - Trained and validated models using standard evaluation practices to assess performance and reliability.
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TECHNICAL SKILLS

Data Analysis & Programming

- Python, SQL, R, C++

Machine Learning & Analytics

- Predictive modeling, feature engineering, model evaluation
- Scikit-learn, Random Forests, regression models
- Multivariate analysis
- Deep Learning & NLP: Large Language Models (LLMs)

Data Visualization & Reporting

- Excel (advanced: large dataset cleaning, pivot tables, lookup functions, conditional formulas, data validation & formatting)
- Power BI (charting, plots, visualizations)

Statistical & Research Tools

- SPSS
- Survey analytics and consumer research data analysis

Soft Skills:

- Analytical thinking and structured problem-solving
 - Clear communication of technical insights
 - Collaboration with cross-functional teams
 - Fast learner with strong adaptability
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